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Designing service-based applications in the presence of non-functional properties: A mapping study

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Abstract

Context

The development of distributed software systems has become an important problem for the software engineering community. Service-based applications are a common solution for this kind of systems. Services provide a uniform mechanism for discovering, integrating and using these resources. In the development of service based applications not only the functionality of services and compositions should be considered, but also conditions in which the system operates. These conditions are called non-functional requirements (NFR). The conformance of applications to NFR is crucial to deliver software that meets the expectations of its users.

Objective

This paper presents the results of a systematic mapping carried out to analyze how NFR have been addressed in the development of service-based applications in the last years, according to different points of view.

Method

Our analysis applies the systematic mapping approach. It focuses on the analysis of publications organized by categories called facets, which are combined to answer specific research questions. The facets compose a classification schema which is part of the contribution and results.

Results

This paper presents our findings on how NFR have been supported in the development of service-based applications by proposing a classification scheme consisting in five facets: (i) programming paradigm (object/service oriented); (ii) contribution (methodology, system, middleware); (iii) software process phase; (iv) technique or mathematical model used for expressing NFR; and (v) the types of NFR addressed by the papers, based on the classification proposed by the ISO/IEC 9126 specification. The results of our systematic mapping are presented as bubble charts that provide a quantitative analysis to show the frequencies of publications for each facet. The paper also proposes a qualitative analysis based on these plots. This analysis discusses how NFR (quality properties) have been addressed in the design and development of service-based applications, including methodologies, languages and tools devised to support different phases of the software process.

Conclusion

This systematic mapping showed that NFR are not fully considered in all software engineering phases for building service based applications. The study also let us conclude that work has been done for providing models and languages for expressing NFR and associated middleware for enforcing them at run time. An important finding is that NFR are not fully considered along all software engineering phases and this opens room for proposing methodologies that fully model NFR. The data collected by our work and used for this systematic mapping are available in https://github.com/placidoneto/systematic-mapping_service-based-app_nfr.

Introduction

We present a systematic mapping (SM) [141] to study the way in which non-functional requirements (NFR) are addressed in the development of service-based applications. SM is a technique for analyzing a field of interest, in our case, service-base software development and NFR. The analysis focuses on the screening of publications, organized by categories called facets and combined to answer specific research questions [27]. The facets are derived by distilling frequent terms from screened papers and using thesauri and vocabularies to identify terms of the related domains of study. These answers are supported by the quantitative data generated through the SM steps.

In systems engineering, NFR, also called *quality properties*, are conditions that refer to the behavior of a system. These conditions are not necessarily related to the output of the system or its application logic, but to the conditions of its execution, its performance, or other properties (such as security, fault tolerance, etc). Associating NFR to service based-applications can help to ensure that the resulting software system complies with the user requirements and with the characteristics of the services it uses.

Diverse terms have been used to refer to NFR. Some of them are “constraints”, “quality attributes”, “quality goals”, “quality of service requirements” or “non-behavioural requirements”. This variability of terms about NFR comes from vocabularies of different domains, like software engineering, distributed systems or service oriented programming. In the case of service-based applications, NFR concern the conditions in which the application is executed as well as the constraints imposed by the services.

The SM presented in this paper aims to identify the evolution of the area between 1998 and 2015, as well as to identify the relationship between concepts used for defining NFR and associating them to service-based applications.

The remainder of this paper is organized as follows. Section2 gives the background about NFR and service-oriented computing. Section3 describes the (standard) SM protocol, and describes and explains our research questions, search string (i.e., query) and papers selection. Section4 presents and interprets the analytic results. Section5 concludes the paper and discusses research perspectives.

Section snippets

Background

The *service-oriented computing*(SOC) [135] paradigm establishes the basis for engineering systems by using services as fundamental software units. In this context, services are autonomous modules conceived and developed by different vendors to offer encapsulated functionalities. Services have well-defined interfaces and provide behaviors according to specific concerns. The service-oriented software process inherits elements and principles from object-orientation analysis and design [64], as well …

Mapping process

For performing our systematic mapping study we adopt the method proposed in [141], consisting of five steps:

1. **Definition of research questions**, to determine the research scope. …

2. **Search of primary papers**, to select candidate papers expressing a query for retrieving references from scientific databases. …

3. **Screening of papers**, to identify relevant papers using inclusion and exclusion criteria to narrow the number of papers of interest. …

4. **Keywording of abstracts**, to identify terms that can be used for …

…

Analysis of the results

The following paragraphs analyze and discuss the results for our systematic mapping study. We proceed by (i) presenting the collected data in a graphic form, in order to discover relationships between existing works and (ii) answering the research questions that have been stated to guide our study. …

Concluding remarks

The SM results presented in this paper can be the starting point to motivate new studies, support the investigation of specific problems not sufficiently explored yet. Our quantitative analysis helps to see the global distribution of the research in the area, as well as the main conferences and journals that publish the results of that research. The qualitative analysis (Section4.1) gives some information about the origin, venue of publication and publishers of the papers that were selected …

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>
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